Science of the Total Environment 627 (2018) 39–45

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Soil nutrient variability and groundwater nitrate-N in agricultural fields☆

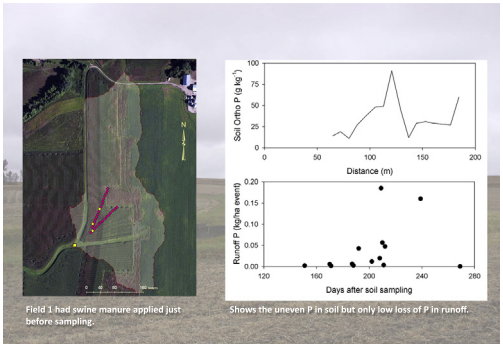
Sally D. Logsdon *, Kevin J. Cole

USDA-ARS, National Laboratory for Agriculture and the Environment, 1015 N. University Blvd., Ames, IA 50011, USA

HIGHLIGHTS

- Significant slope or curvature effects only for large range
- No significant landscape effects for large nutrient range
- High groundwater nitrate-N low areas: additions balanced losses

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 1 December 2017
Received in revised form 16 January 2018
Accepted 18 January 2018
Available online xxxx

Editor: Jay Gan

Keywords:
Groundwater nitrate
Landscape
Soil phosphorus

ABSTRACT

Landscape may result in uneven nutrient loads within a field. The objective of this study was to determine the influence of landscape on soil carbon and nutrient levels, and on levels of nitrate-N in groundwater. Soil samples were collected in three fields, two transects each, 30 sites in each field. The soil morphology was characterized for the profile, and soil organic carbon and nutrient levels were determined for 0–0.15 and 0.15–0.3 m depths. Each field had wells installed at three of the sites. One field showed a wide range of landscape variability, and significant effects of curvature on soil carbon and nutrient levels. Another field showed no significant effect of slope or curvature on soil carbon and nutrient levels because the nutrient levels were quite variable, including high spikes. The third field had less variable landscape trends but still showed a few significant effects on soil carbon and nutrient levels. Nitrate-N levels remained high in two of the nine wells (20 to 50 mg L^{-1}), suggesting that additions of nitrate-N at the concave or converging sites replaced any losses. Median nitrate-N levels at the other seven well sites were lower, ranging from 8 to 17 mg L^{-1}. Influence of landscape on soil carbon and nutrients was more detectable when the landscape factors were highly variable without excessive variability in soil nutrient properties.

Published by Elsevier B.V.

1. Introduction

Phosphorus losses from agricultural fields are primarily in runoff, whereas nitrate is lost to groundwater (Pärn et al., 2012). Landscape and management may result in uneven nutrient loads within field soils. Water tends to accumulate in low slope, concave, and converging parts of the landscape; however, high slope, convex, and diverging regions usually shed water (Moore et al., 1988). Erosion due to water and tillage may transport nutrient-laden surface soil from shoulder and backslope positions to footslope and toeslope positions (Kaspar et al., 2004; Priyashantha et al., 2007; Li et al., 2013). Sacchi et al. (2013) showed higher nitrate-N levels in groundwater at footslope sites than for other landscape positions. On the other hand, ponding

☆ This research was funded from base funds for USDA-ARS CRIS 5030-1300-010-00D, Accession 0422830.

* Corresponding author.
E-mail address: sally.logsdon@ars.usda.gov (S.D. Logsdon).

https://doi.org/10.1016/j.scitotenv.2018.01.182
0048-9697/Published by Elsevier B.V.

within depressions could contribute to denitrification or leaching of nitrate (Cambardella et al., 1999). Footslope and toeslope sites often have higher levels of soil organic carbon (Ritchie et al., 2007; Li et al., 2013; Fissore et al., 2017) than shoulder or backslope positions.

The footslope, toeslope, and depressional areas with high nutrient levels in the surface soil may be located close to tiles, ditches, and grass waterways in fields. Depressions and low slope areas can trap nutrient- and carbon-rich sediment (Beach, 1994; Almendinger et al., 2014). Eroded sediments tend to be deposited close to the eroded site, and the newly deposited sediments could be a source for future erosion (VandenBygaart et al., 2012).

The hypotheses of this study were that 1) soil organic carbon and phosphorus levels are higher in low slope, concave, and converging parts of the landscape, and 2) groundwater nitrate-N concentrations at lower landscape positions will be decreased. The objective of this study were to determine if landscape factors (slope and curvature) influenced carbon and nutrient levels in surface soil or nitrate levels in groundwater.

2. Materials and methods

2.1. Site descriptions

The sites examined (Fig. 1) included Des Moines lobe sites: a field in Walnut Creek watershed (Hatfield et al., 1999) and two fields in the South Fork watershed of the Iowa River (Tomer et al., 2008). The Des Moines lobe is characterized by limited stream development, mainly at the lower outlet end of watersheds, with many closed depressions upland. Now the area is extensively tile-drained.

Field 1 (Figs. 2, and S1) in the South Fork watershed was close to the eastern edge of the Des Moines lobe and had the steepest slopes. The latitude was 42°19′28.3″N, and longitude was 93°15′20.2″W. There were fall additions of swine manure before corn (*Zea mays* L.) the following season. The injected manure was disked in (Tomer et al., 2016). The rotation was corn - corn - soybean (*Glycine* max (L.) Merr), with soybean in 2012 and 2015. This site had a flume for runoff monitoring (Tomer et al., 2016).

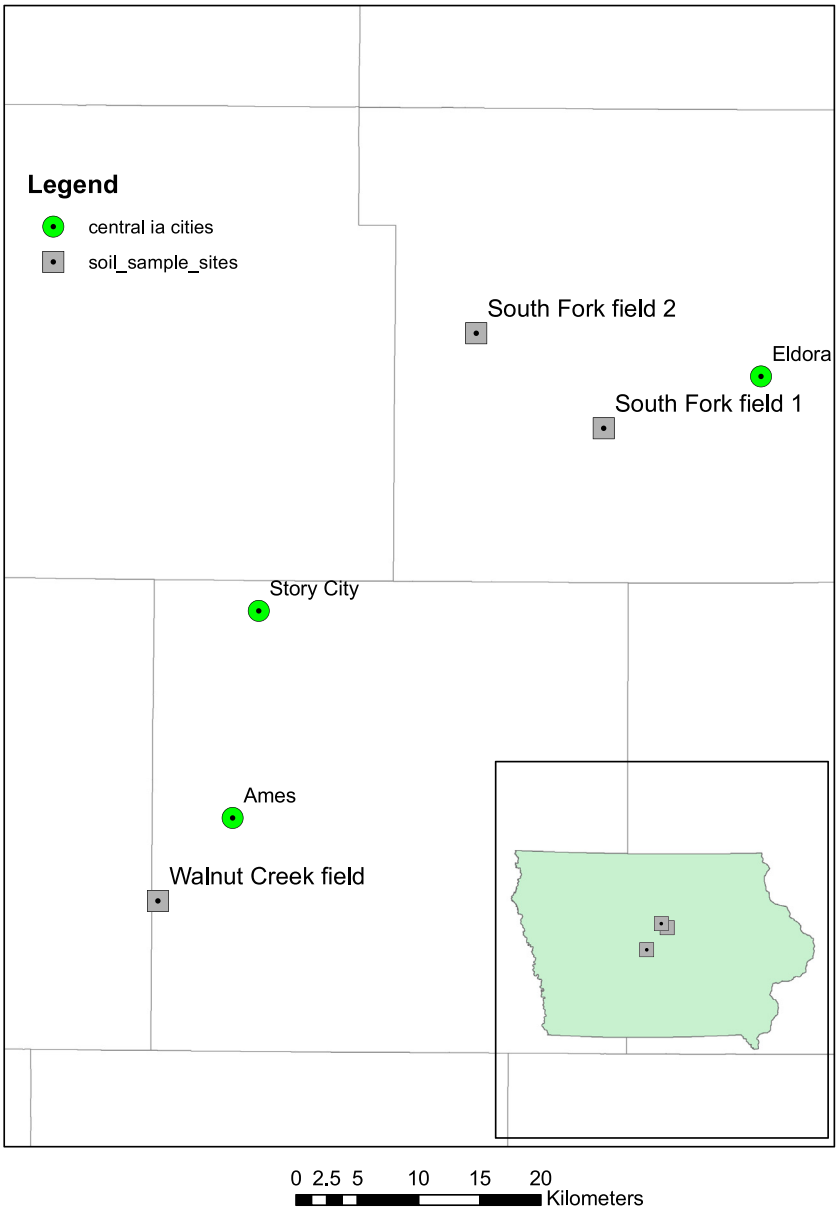


Fig. 1. Location of the three sampled fields in Iowa.

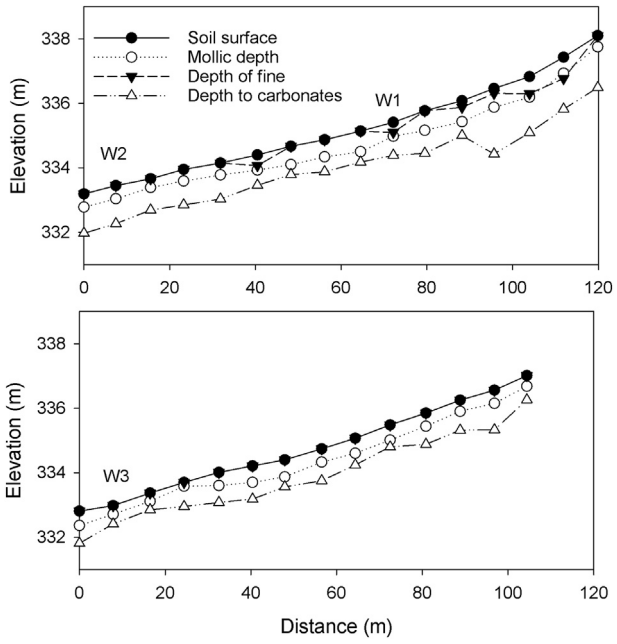


Fig. 2. Field 1 in South Fork showing depth of Mollic epipedon, depth of fine glacial deposits, and depth to carbonates for transect 1 (upper) and 2 (lower). Locations of well 1 (W1), 2 (W2), and 3 (W3) are indicated.

Field 2 (Figs. 3 and S2) in the South Fork watershed was more centrally-located within the watershed. The latitude was 42°23′32.0″ N, and longitude was 93°22′56.1″W. Each spring, cattle manure with bedding was applied lightly and mixed with the soil surface. In 2015 the farm manager changed. This was a corn - corn - soybean rotation through 2014 with soybean in 2013. The new farm manager changed to continuous corn with residue removal.

The field in Walnut Creek (Figs. 4 and S3) watershed was the most level site. The latitude was 41°58′24.5″N, and is −93°41′35.5″W. This was a corn - soybean rotation with corn in 2014. All fields had fertilizer and anhydrous ammonia additions for corn.

2.2. Sample and site properties

In the fall of 2012 (a drought year) at each field, 30 soil cores were collected in plastic tubes ~5 cm inner diameter along topographic transects, extending to >1.5 m depth (up to 2.5 m if possible) using a hydraulic probe. The samples were used for describing thickness of mollic epipedon, depth to carbonates, and depth of fine-textured material at the soil surface. Soil texture was determined "by feel" (Thien, 1979); however, samples with extensive sand were dispersed and dried for sand fractionation (but not silt-clay separation). Soil 0–0.15 and 0.15–0.30 m were used to determine ortho phosphorus (P) with the Olsen extraction (Olsen et al., 1954) for colorimetric phosphate determination. Also nitrate and ammonium were determined from these samples after extraction by 2 M KCl with a Lachet autoanalyzer that used Cd reduction (Wood et al., 1967) using flow injection technology (Lachet Instruments, Milwaukee, WI). Total carbon and nitrogen were determined by dry combustion (Nelson and Sommers, 1982) with a Fisons NA1500 Elemental Analyzer. A few surface samples had carbonates, so inorganic carbon (Sherrod et al., 2002) was determined for these samples to get organic carbon by subtraction.

A 1-m grid digital elevation model (James and Gelder, personal communication, 2012) derived from a LiDAR survey was used to determine slope, profile and plan curvatures by a python program. The three-by-three neighborhood used the points 10 m out rather than 1 m; however, the data were not resampled, and the terrain parameters were determined for all points (Logsdon and James, 2014).

For each field, the top fourth quartile distribution was determined for slope to compare with the bottom three quartiles distribution. For profile and plan curvatures, the bottom quartiles distribution (either more convex or more divergent) was compared with the mid quartiles (linear) and top quartile distribution (either more concave or more convergent). Landscape differences were tested with a non-parametric rank test due to non-normality of some data. The Mann-Whitney rang test was used for slope comparisons (two groups) and the Kruskal-Wallis analysis for the curvatures (three groups each).

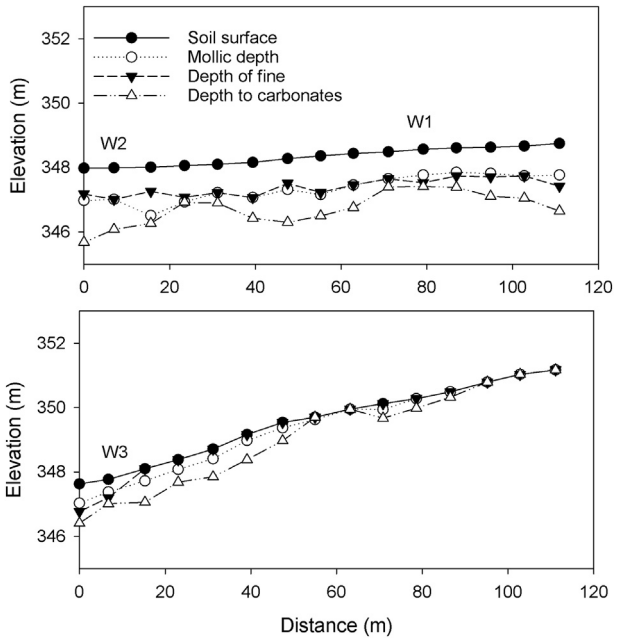


Fig. 3. Field 2 in South Fork showing depth of Mollic epipedon, depth of fine glacial deposits, and depth to carbonates for transect 1 (upper) and 2 (lower). Locations of well 1 (W1), 2 (W2), and 3 (W3) are indicated.

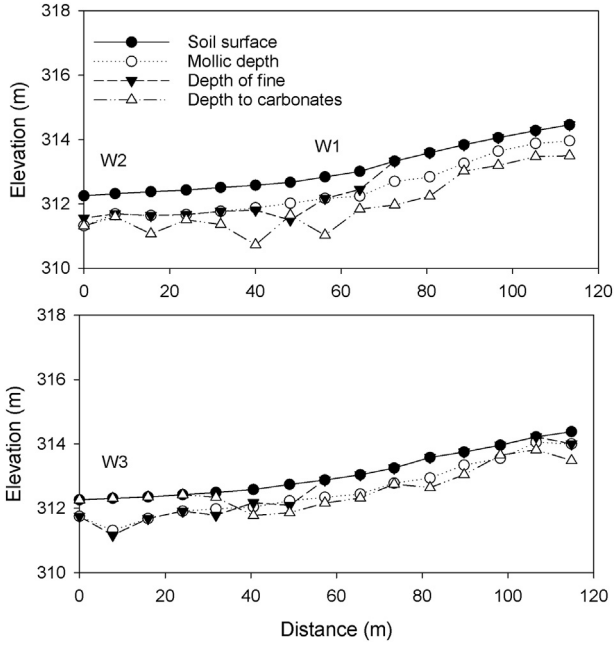


Fig. 4. The field 1 in Walnut Creek showing depth of Mollic epipedon, depth of fine glacial deposits, and depth to carbonates for transect 1 (upper) and 2 (lower). Locations of well 1 (W1), 2 (W2), and 3 (W3) are indicated.

2.3. Water samples

For each field, wells were installed at three of the sites (Table 1, Figs. 2 to 4 and S1 to S3). They were sampled monthly for nitrate-N (unless the wells were dry or snow drifts prevented field entry). The well samples were analyzed for nitrate plus nitrite nitrogen with a Lachet autoanalyzer that used Cd reductions (Wood et al., 1967; Tomer et al., 2008). We determined distance from each well site to the nearby grassed waterways for the two fields on South Fork watershed. Fields 1 and 2 of the South Fork watershed were close to grass waterways, so distance to the grass waterways was measured for each of the well sites. Also there were tiles in the grassed waterways, assumed to be in the lowest part of the waterway, allowing an estimate of distance to assumed tile placement (actual tile location not verified). The well sites for field in the Walnut Creek watershed were not close to tiles, but distance was determined to the closest tile from tile maps (actual tile location not verified).

Runoff data, for 2013 in field 1 of South Fork watershed, was continued based on the procedure described in Tomer et al. (2016), using an H-type flume. Total P concentrations were determined using flow injection analysis (US EPA, 1993) after acid-persulfate digestion. P loads were determined for field 1 based on the 7.6 ha cropped area, as discussed in Tomer et al. (2016). Runoff was not collected in the other two fields.

3. Results

3.1. Soil morphology

Field 1 of South Fork watershed (Fig. 2, Table 2) was steeper than the other sites, and there were some sub-surface loamy sand layers. Most of the sites in field 1 were Clarion (Fine-loamy, mixed, superactive, mesic Typic Hapludoll) with a small area of Terrill (Fine-loamy, mixed, superactive, mesic Cumulic Hapludoll) near the top of transect 1 (Soil Survey Staff, 2014). All of transect 1 was high slopes, but the upper part of the transect was concave and had some fine sediment at the soil surface (Fig. 3, Table S1). Across the two transects (Table S2), 11 out of 30 sites were diverging.

Field 2 of South Fork watershed (Fig. 3, Table 3) had one transect mostly converging (Table S3) and the other mostly diverging (Table S4), hence more variability than the other fields. The convergent transect 1 had deep mollic epipedons and zones of sub-surface loamy sand. There were also narrow zones of either buried A horizons or krotovinas. The divergent transect 2 had many sites in which A horizons did not meet the criteria for a mollic epipedon (not dark enough or too shallow). Most of transect 2 was Clarion (with mollic epipedon) or Storden (no mollic epipedon: Fine-loamy, mixed, superactive, mesic Typic Eutrudept). The Storden areas also had carbonates at the soil surface due to reduced leaching and excess erosion. The lowest two sites for transect 2 were in the low area with soils as in transect 1, generally Coland (fine-loamy, mixed, superactive, mesic Cumulic Endoaquoll) - Terrill with some areas of Webster (Fine-loamy, mixed, superactive, mesic Typic Endoaquoll) -Nicollet.

The field in Walnut Creek watershed (Fig. 4, Table 4) was the flattest. Several sampling points had either buried A horizons or krotovinas, and subsurface loamy sand and gravel layers. In this field, transect 1 (Table S5) had surface carbonate accumulation near a depression, but the depression itself was not included in the transect since it was located outside the field. Around depressions there is upward movement of carbonates (Miller et al., 1985; Arndt and Richardson, 1989; Logsdon and James, 2014). The upper parts of both transects (Table S6) was Clarion, and the middle and bottom sites were Webster with some Canisteo (Fine-loamy, mixed, superactive, calcareous, mesic Typic Endoaquoll) and Harps (Fine-loamy, mixed superactive, mesic Typic Calciaquoll) for the bottom of transect 1.

3.2. Soil carbon and nutrient levels in relation to slope and curvature

Field 1 in the South Fork watershed did not show any significant relation of soil nutrients or carbon to slope or curvatures because there was large variability in soil nutrients (Tables 2 and S7). A high spike in nutrient concentration was located 20 m from the grassed waterway (at well 1 site, Fig. 2), possibly due to uneven swine manure application a couple weeks before sampling. Unfortunately, we did not know until after sampling about the nutrient spike, so no pre-application soil samples were taken to verify that manure application caused the variation.

Table 1
Properties at well sites in the three fields.

Site	Mollic	Buried dark	Depth fine	Sand	Dist. waterway	Dist. tile	Land position
	m	m	m	m	m	m	
SF1 well 1	0.53	–	0	0.88–1.04	20	25	TS
SF1 well 2	0.41	–	0	–	7	12	TS
SF1 well 3	0.27	–	0	–	16	21	TS-c
SF2 well 1	0.67	1.48–1.88	1.04	1.65–1.98	52	7	TS
SF2 well 2	0.98	1.17–1.26, 1.83–1.91	0.98	1.26–2.20	16	7	TS-c
SF2 well 3	0.39	–	0.76	–	7	7	FS
WC well1	0.67	–	0.67	0.90–1.65, 1.71–1.73	–	83	TS-c
WC well 2	0.63	–	0.63	0.72–0.90	–	42	TS-c
WC well 3	0.82	1.15–1.26	0.82	1.26–1.60	–	24	TS

SF refers the fields 1 and 2 in the South Fork watershed, and WC refers to the field in Walnut.
Creek watershed. Dist. is distance to waterway or estimated to tile. BS is back slope, FS is footslope, TS is toeslope, c is converging.

Table 2
South Fork field 1 transect soil properties: minimum, first quartile, median, third quartile, maximum, for 0–0.15 and 0.15–0.3 m depths.

Property	Min	25	Med	75	Max
NO_3 0.0–0.15 mg kg^{-1}	4.2	8.0	13.8	26.6	136.5
NO_3 0.15–0.3 mg kg^{-1}	3.6	4.4	5.3	8.0	40.1
NH_4 0.0–0.15 mg kg^{-1}	3.1	4.1	5.2	7.6	124.3
NH_4 0.15–0.3 mg kg^{-1}	1.8	2.4	3.0	4.7	5.7
Ortho P 0.0–0.15 mg kg^{-1}	8	14	23	31	91
Ortho P 0.15–0.3 mg kg^{-1}	1	2	3	5	9
OC 0.0–0.15 g per 100 g	1.30	1.62	1.94	2.06	2.45
OC 0.15–0.3 g per 100 g	1.01	1.29	1.54	1.70	2.08
TN 0.0–0.15 g per 100 g	0.13	0.15	0.17	0.19	0.21
TN 0.15–0.3 g per 100 g	0.11	0.13	0.14	0.15	0.17
Slope (m per 100 m)	2.97	3.685	4.31	6.47	14.56
Profile curvature (m^{-1})	−0.140	0.049	0.085	0.147	0.378
Plan curvature (m^{-1})	−0.493	−0.096	−0.041	0.020	0.168

OC is organic carbon, and TN is total nitrogen.

Table 3
South Fork field 2 transect soil properties: minimum, first quartile, median, third quartile, maximum, for 0–0.15 and 0.15–0.3 m depths.

Property	Min	25	Med	75	Max
NO_3 0.0–0.15 mg kg^{-1}	4.1	5.2	10.0	16.0	32.1
NO_3 0.15–0.3 mg kg^{-1}	2.1	3.1	5.2	8.8	19.2
NH_4 0.0–0.15 mg kg^{-1}	2.0	3.0	3.6	4.1	6.1
NH_4 0.15–0.3 mg kg^{-1}	1.1	3.0	3.6	4.1	6.1
Ortho P 0.0–0.15 mg kg^{-1}	1	2	6.5	21	29
Ortho P 0.15–0.3 kg^{-1}	0	0	1.5	4	10
OC 0.0–0.15 g per 100 g	1.21	1.64	2.40	2.92	3.48
OC 0.15–0.3 g per 100 g	0.42	0.79	2.18	2.72	3.20
TN 0.0–0.15 g per 100 g	0.12	0.14	0.20	0.24	0.30
TN 0.15–0.3 g per 100 g	0.06	0.09	0.18	0.21	0.26
Slope (m per 100 m)	0.21	1.59	2.25	3.99	6.48
Profile curvature (m^{-1})	−1.134	−0.103	0.113	0.195	0.412
Plan curvature (m^{-1})	−0.677	−0.305	−0.022	0.083	0.242

OC is organic carbon, and TN is total nitrogen.

Table 4
Walnut Creek field transect soil properties: minimum, first quartile, median, third quartile, maximum, for 0–0.15 and 0.15–0.3 m depths.

Property	Min	25	Med	75	Max
NO_3 0.0–0.15 mg kg^{-1}	5.8	9.5	11.0	16.7	36.7
NO_3 0.15–0.3 mg kg^{-1}	3.1	4.4	6.5	10.1	22.2
NH_4 0.0–0.15 mg kg^{-1}	2.7	4.0	5.1	5.5	7.5
NH_4 0.15–0.3 mg kg^{-1}	1.1	2.0	2.7	3.2	7.5
Ortho P 0.0–0.15 mg kg^{-1}	3	5	10.5	18	33
Ortho P 0.15–0.3 mg kg^{-1}	2	2	3	3	12
OC 0.0–0.15 g per 100 g	1.34	2.18	3.59	4.21	4.88
OC 0.15–0.3 g per 100 g	1.11	1.68	2.46	3.25	4.19
TN 0.0–0.15 g per 100 g	0.12	0.19	0.29	0.34	0.38
TN 0.15–0.3 g per 100 g	0.12	0.15	0.18	0.25	0.28
Slope (m per 100 m)	0.50	1.00	2.09	3.37	3.75
Profile curvature (m^{-1})	−0.274	−0.010	0.0175	0.049	0.242
Plan curvature (m^{-1})	−0.187	−0.053	−0.011	0.0155	0.065

OC is organic carbon, and TN is total nitrogen.

Field 2 of South Fork watershed included erosional sample points in transect 2 combined with depositional points in transect 1, which perhaps showed a broader range of soil conditions than for the other two fields. This may have allowed detection of significant effects of slope and curvature on soil nutrient and carbon levels for this field (Table 5). Profile and plan curvatures significantly affected soil nutrient and carbon levels more than slope affected these soil properties. Concave or concave and linear profile curvature often had significantly higher nutrient and carbon amounts than convex sites. Likewise, converging or converging and linear plan curvature often had significantly higher nutrient and carbon amounts than diverging sites.

Table 5
Slope and curvature effects on soil nutrients for field 2 in South Fork watershed.

Property	Slope > 3.99 m per 100 m	Slope < 3.99 m per 100 m
NO_3 0.0–0.15 mg kg^{-1}	6.3a	13.6a
NO_3 0.15–0.3 mg kg^{-1}	3.4a	5.7a
NH_4 0.0–0.15 mg kg^{-1}	2.0b	3.9a
NH_4 0.15–0.3 mg kg^{-1}	1.5b	2.1a
Ortho P 0.0–0.15 mg kg^{-1}	2b	14a
Ortho P 0.15–0.3 mg kg^{-1}	0b	3a
OC 0.0–0.15 g per 100 g	1.85a	2.68a
OC 0.15–0.3 g per 100 g	0.96b	2.48a
TN 0.0–0.15 g per 100 g	0.16b	0.23a
TN 0.15–0.3 g per 100 g	0.10b	0.20a

Property	Prof. Curve. > 0.195	Lin.	Prof. Curve<−0.102
NO_3 0.0–0.15 mg kg^{-1}	16.0a	10.3b	4.6b
NO_3 0.15–0.3 mg kg^{-1}	7.9a	5.9b	3.1b
NH_4 0.0–0.15 mg kg^{-1}	3.9a	4.0a	3.1a
NH_4 0.15–0.3 mg kg^{-1}	2.1a	2.2a	1.5b
Ortho P 0.0–0.15 mg kg^{-1}	15a	11.5b	2c
Ortho P 0.15–0.3 mg kg^{-1}	2ab	3a	NDb
OC 0.0–0.15 g per 100 g	2.79a	2.60a	1.43b
OC 0.15–0.3 g per 100 g	2.64a	2.34a	0.69b
TN 0.0–0.15 g per 100 g	0.23a	0.22a	0.14b
TN 0.15–0.3 g per 100 g	0.21a	0.19a	0.09b

Property	Plan Curve. > 0.083	Lin.	Plan Curve<−0.305
NO3 0.0–0.15 mg kg^{-1}	16.0a	10.0ab	4.7b
NO3 0.15–0.3 mg kg^{-1}	1.24a	5.2b	2.7c
NH4 0.0–0.15 mg kg^{-1}	3.9a	3.6a	3.1a
NH4 0.15–0.3 mg kg^{-1}	2.1a	2.1a	1.5b
Ortho P 0.0–0.15 mg kg^{-1}	21a	9.5a	3a
Ortho P 0.15–0.3 mg kg^{-1}	4a	2ab	0b
OC 0.0–0.15 g per 100 g	3.13a	2.47a	1.36b
OC 0.15-.3 mg per 100 g	2.99a	2.19b	0.69c
TN 0.0–0.15 g per 100 g	0.26a	0.20b	0.14c
TN 0.15–0.3 mg per 100 g	0.22a	0.17ab	0.09b

Medians followed by the same letter within a row are not significantly different at p = 0.05.
OC is organic carbon, and TN is total nitrogen.
ND means none was detected.

There were a few significant trends for the Walnut Creek field (Table S8). Soil nitrate 0–15 cm was significantly lower for the slopes in the upper quadrant (10.5) than the other three quadrants (14.1). Organic carbon 0–15 cm was significantly higher (3.9 g per 100 g) for the linear profile curvature sites than for the convex sites (2.0); however, the linear plan curvature sites had significantly less (2.5) organic carbon than the converging (4.1) or diverging (4.2) sites. Organic carbon for the 15–30 cm depth the linear profile curvature had significantly higher organic carbon levels (3.0) than the convex sits (1.7), and the linear plan curvature sites had significantly less (1.7) than the converging (3.1) or diverging (3.3) sites. Total nitrogen 0–15 cm was also significantly lower (0.17) for the plan linear sites than for the converging (0.31) or diverging (0.3) sites. Other properties were not significantly affected by slope or curvature at the Walnut Creek field, perhaps due to lower variability of landscape features in these transects.

3.3. Phosphorus in runoff

In this study, swine manure had been applied to field 1 in South Fork watershed in October 2012 when the soil was quite dry. The soil was sampled on 3 November 2012. There were a few snow-melt runoff events in February or March 2013, but the first rain-driven runoff event was not until April 2013. Runoff levels per event were <20 mm (Fig. 5), and P loads per event remained <0.2 kg ha^{-1}.

3.4. Groundwater nitrate-N

Well sample nitrate-N concentrations from well 1 in field 1 of the South Fork watershed (Fig. 6 top, Fig. S4) were highest (24 mg L^{-1}) on 7/29/2013 when the water table had risen to 1.15 m after a dry season 2012. The lowest nitrate-N concentrations (8 mg L^{-1}) occurred on 9/22/2015 when the water table was at 1.11 m after wetter seasons in 2014 and 2015. A similar trend was observed for well 2 with highest values (19 mg L^{-1}) on 7/29/2013 when the water table was at 1.18 m, and lowest values (10 mg L^{-1}) on 9/22/2015 when the water table was at 1.00 m. Well 3 showed the highest value (21 mg L^{-1}) on 6/28/2013 when the water table was at 0.61 m, and the lowest value (1 mg L^{-1}) on 8/27/2013 when the water table was at 2.56 m. Median nitrate-N concentrations across all measurement dates were 17.2, 15.4, and 10.1 mg L^{-1} for wells 1, 2, and 3.

Well nitrate-N concentrations were high for two of the wells in field 2 of South Fork watershed (Fig. 6 mid, Fig. S5). The highest value (30 mg L^{-1}) for well 1 occurred on 6/28/2013 when the

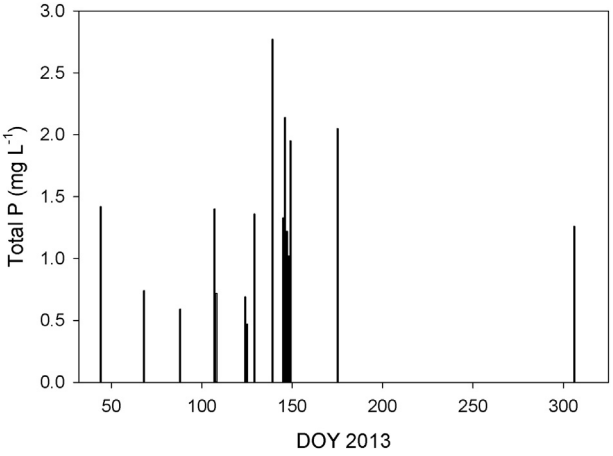


Fig. 5. Total P concentration in runoff events for 2013, following manure application in October 2012 for field 1 in South Fork watershed.

water table depth was 1.2 m, and the lowest value (6 mg L^{-1}) occurred on 9/26/2013 when the water table depth was 3.03 (almost a dry well). For well 2, the highest nitrate-N value (47 mg L^{-1}) occurred on 7/31/2014 when the water table depth was 1.13 m. The lowest nitrate-N value (15 mg L^{-1}) occurred on 9/21/2015 when the water table was at 1.05 m. Values were also high for well 3 with the highest value (50 mg L^{-1}) occurring on 5/23/2014 when the water table depth was 0.82 m. The lowest value (23 mg L^{-1}) occurred on 9/21/2015 when the water table was at 1.0 m. Median nitrate-N values were 14.4, 30.9, and 40.8 mg L^{-1} for wells 1, 2, and 3.

Well nitrate-N concentrations were lower for the Walnut Creek field (Fig. 6 bottom) due to higher water tables (Fig. S6) caused by more rain and great distance to buried tiles. The highest nitrate-N concentrations (23 mg L^{-1}) for well 1 was on 5/06/2014 when the water table depth was 0.69 m, and lowest concentrations (3) occurred on 7/15/2015 when the water table depth was 0.57 m. The highest level for well 2 (23 mg L^{-1}) occurred on 5/06/2014 when the water table was at 0.49 m, and the lowest amount was (1 mg L^{-1}) on 9/26/2013 when the water table was at 1.96 m. Well 3 showed the highest amount (25 mg L^{-1}) on 7/31/2014 when the water table depth was at 0.22 m deep, and the lowest concentration (1 mg L^{-1}) on 8/27/2013 when the water table depth was 1.6 m. Median nitrate-N values were 12.6, 11.2, and 8.0 for wells 1, 2, and 3.

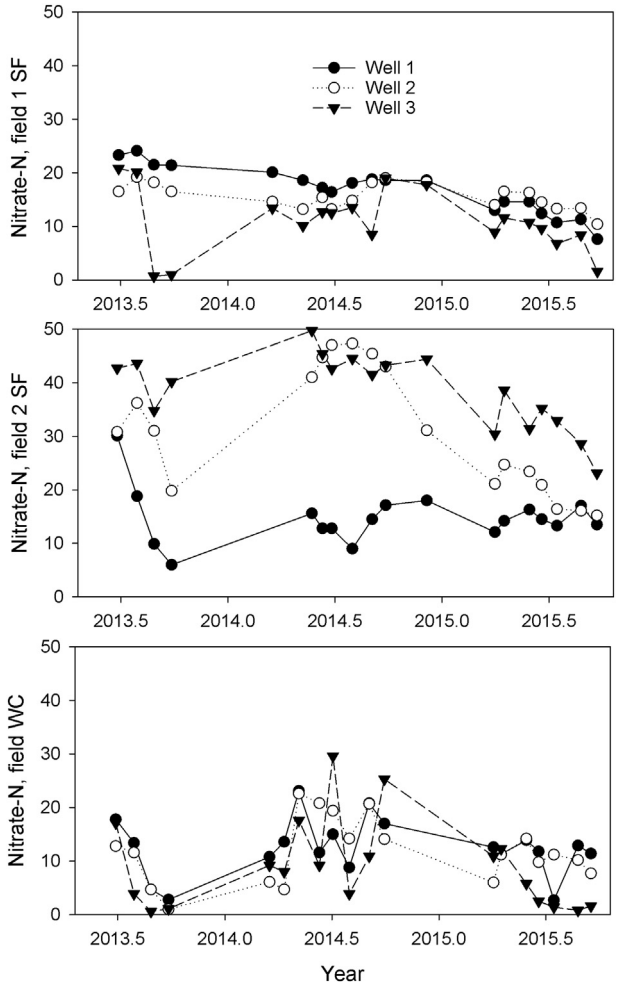


Fig. 6. Nitrate-N concentrations in groundwater for three wells in each field: field 1 of South Fork watershed (SF), field 2 of South Fork watershed, and the field from Walnut Creek watershed (WC).

4. Discussion

4.1. Soil organic carbon and nutrient landscape variability

Soil organic carbon and nutrients were related to slope and curvatures for field 2 of the South Fork watershed (Table 5) due to a broader range of slopes and curvatures. There were no significant relations for field 1 of the South Fork watershed (Table S7), which had greater variability of nutrient levels than landscape factors. Others have shown relations of landscape to soil nutrients or organic carbon. Kaspar et al. (2004) observed greater P in depressional soils than elsewhere on the landscape. On a manure treated field, Priyashantha et al. (2007) showed that surface (0–50 mm) soil nitrate levels remained high at footslope positions perhaps due to runon from upslope, whereas nitrate levels declined at shoulder and backslope positions. Li et al. (2013) showed that toeslope positions had significantly higher levels of soil phosphorus and organic carbon than shoulder or backslope positions. Ritchie et al. (2007) showed that in the top 0.3 m, organic carbon was higher for concave than convex sites, and for converging than for diverging sites. Fissore et al. (2017) showed that concave sites had significantly more total carbon stock in the profile than convex sites, and converging sites had significantly more than linear plan curvature sites.

4.2. Runoff P

In the current study, loss of soil phosphorus in runoff (field 1 in South Fork watershed) was low even though soil levels were high (Fig. 5, Table 2). Soil ammonium and phosphate were considerably higher in the 0–0.15 m depth than the 0.15–0.3 m depth (Table 2). The injected liquid swine manure was disked in, but effective disk depth is <0.15 m (Staricka et al., 1991). Fang et al. (2002) suggest that Olsen-extracted P should be <40 to 55 mg kg^{-1} to prevent runoff losses. Two of the thirty sites (0–0.15 m depth) at field 1 of South Fork watershed exceeded 55 mg kg^{-1} and six exceeded 40 mg kg^{-1}. The highest nutrient spike was coincidently located at well 1 (Fig. 2). When averaged over the whole 0–0.3 m depth-interval, only one sample was >40 mg kg^{-1}. Mean P concentration 0–0.3 m was 16 mg kg^{-1}, or in the high nutrient range (Mallarino et al., 2000). Seven of the 30 samples were >21 mg kg^{-1}, in the very high range.

Tomer et al. (2016) summarized 11 years of previous data from this monitoring site. In their study, 60% of P was lost in the three months following manure application in early May (delayed due to a wet fall). Gonzalez-Hidalgo et al. (2013) has shown that across many sites and years, a few major events account for most of the soil loss. Tomer et al. (2016) considered runoff <20 mm to be low, yet all of the 2013 runoff events in the current study were <20 mm. Tomer et al. (2016) at this site showed the 50th percentile of P load at 0.031 kg P ha^{-1} $event^{-1}$. For the current study, only five of the 13 events in 2013 showed P loads higher than 0.031 kg P ha^{-1} $event^{-1}$, and only two were higher than 0.16 kg P ha^{-1} $event^{-1}$ (26 May and 24 June 2013). Allen and Mallarino (2008) showed that if excess rainfall were delayed for six months after swine manure application, the P concentrations and total loads in runoff would not be significantly different from sites without manure applications. For field 1 of South Fork watershed, the first runoff event was due to snow melt three months after manure application. The first rainfall-runoff event was over four months after manure application.

4.3. Groundwater N

Field 2 of the South Fork watershed (Fig. 6) showed that two of the three (wells 2 and 3) had continual high groundwater nitrate-N levels (range of 15 to 50 mg kg^{-1}) over time (season and off-season) and weather conditions (high and low water table depths). Well 2 was at a converging site and well 3 was at a concave (footslope) site (Table 1), which can contribute to sub-surface lateral flow. Lateral

flow is common in the gently-rolling Des Moines lobe landscape (Logsdon, 2007). Sacchi et al. (2013) showed higher nitrate-N levels in groundwater at footslope sites than for other landscape positions. For the current study, wells 2 and 3 of field 2 were nearest to the grassed waterway. Robertson and Schiff (2008) showed continued high nitrate-N levels in a riparian groundwater zone due to inadequate organic carbon for denitrification. For the current study, the groundwater at well 1 did show some decline in nitrate-N levels over the season (half of the time < 14.4 mg kg^{-1} nitrate-N). Dark material was present at 1.65–1.88 m for well 1 (possibly krotovina), which could have provided a carbon source for denitrification, especially in 2014 when the water table was higher than the 1.65 m depth.

Surface manure applications can result in nitrate in the soil profile. Turkeltaub et al. (2016) used isotopes to show that most nitrate deeper in the soil profile was from manure applications. Field 1 in South Fork watershed had additions of liquid swine manure and field 2 in South Fork watershed had applications solid beef manure with bedding. The Walnut Creek field had no manure additions and lower nitrate-N concentrations (1 to 30 mg kg^{-1}), similar to the range reported by Cambardella et al. (1999) for a different field in the Walnut Creek watershed (0 to 24 mg kg^{-1}). The field in the Walnut Creek watershed had higher water table depths than the other fields (Figs. S4 to S6), which also may have contributed to greater nitrate loss from this field.

5. Conclusions

Field 2 of the South Fork watershed (more variable landscape positions) showed numerous significant relationships between slope and curvatures and soil organic carbon and nutrient levels. Field 1 of the South Fork watershed did not show these relationships because the nutrient levels were high and variable. The high nutrient levels in field 1 of South Fork watershed did not result in much loss of phosphorus in runoff, because runoff was delayed after these high nutrient levels had been measured. So the first hypothesis that landscape influenced soil carbon and nutrient levels was only partially upheld by the data.

Nitrate-N levels in groundwater were consistently high for a converging site and for a concave site in field 2 of the South Fork watershed. This was inconsistent with the hypothesis of lower nitrate-N levels for lower landscape positions. Perhaps the rate of nitrate-N lateral addition exceeded any loss by denitrification or leaching at these sites.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2018.01.182.

Conflict of interest statement

We declare that we have no conflict of interest with other people or organizations that could have influenced the content of the manuscript.

Acknowledgements

Thanks to Gavin Simmons for collection of field data. Thanks to the three farmers for permitting data collection in their fields. Thanks to Mark Tomer and Keith Schilling for reviewing an earlier version of this manuscript. Thanks to Michael Marti for refining the graphical abstract. USDA is an equal opportunity provider and employer. Thanks to the reviewers for helpful suggestions.

References

Allen, B.L., Mallarino, A.P., 2008. Effect of liquid swine manure rate, incorporation, and timing of rainfall on phosphorus loss with surface runoff. J. Environ. Qual. 37: 125–137. https://doi.org/10.2134/jeq2007.0125.

Almendinger, J.E., Murphy, M.S., Ulrich, J.S., 2014. Use of the soil water assessment tool to scale sediment delivery from field to watershed in an agricultural landscape with topographic depressions. J. Environ. Qual. 41:9–17. https://doi.org/10.2134/jeq2011.0340.

Arndt, J.L., Richardson, J.L., 1989. Geochemistry of hydric soil salinity in a recharge-throughflow-discharge prairie-pothole wetland system. Soil Sci. Soc. Am. J. 53: 848–855. https://doi.org/10.2136/sssaj1989.03615995005300030037x.

Beach, T., 1994. The fate of eroded soil: sediment sinks and sediment budgets of agrarian landscapes in southern Minnesota, 1851–1988. Ann. Assoc. Am. Geogr. 84, 5–28.

Cambardella, C.A., Moorman, T.B., Jaynes, D.B., Hatfield, J.L., Parkin, T.B., Simpkins, W.W., Karlen, D.L., 1999. Water quality in Walnut Creek watershed: nitrate-nitrogen in soil subsurface drainage water, and shallow groundwater. J. Environ. Qual. 28, 25–34.

Fang, F., Brezonik, P.L., Mulla, D.J., Hatch, L.K., 2002. Estimating runoff phosphorus losses from calcareous soils in the Minnesota River basin. J. Environ. Qual. 31, 1918–1929.

Fissore, C., Dalzell, B.J., Berhe, A.A., Voegtle, M., Evans, M., Wu, A., 2017. Influence of topography on soil organic carbon dynamics in a Southern California grassland. Catena 149:140–149. https://doi.org/10.1016/j.catena.2016.09.016.

Gonzalez-Hidalgo, J.C., Batalla, B.J., Cerda, A., 2013. Catchment scale and contribution of the largest daily events to suspended sediment load on a continental scale. Catena 102:40–45. https://doi.org/10.2134/jeq2012.0233.

Hatfield, J.L., Jaynes, D.B., Burkart, M.R., Cambardella, C.A., Moorman, T.B., Prueger, J.H., Smith, M.A., 1999. Water quality in Walnut Creek watershed: setting and farming practices. J. Environ. Qual. 23, 11–24.

Kaspar, T.C., Pulido, D.J., Fenton, T.E., Colvin, T.S., Karlen, D.L., Jaynes, D.B., Meek, D.W., 2004. Relationship of corn and soybean yield to soil and terrain properties. Agron. J. 96, 700–709.

Li, F., Zhang, J., Su, Z., Fan, H., 2013. Simulation and Cs-137 tracer show tillage erosion translocating soil organic carbon, phosphorus, and potassium. J. Plant Nutr. Soil Sci. 176:647–654. https://doi.org/10.1002/jpln.201200341.

Logsdon, S.D., 2007. Subsurface lateral transport in glacial till soils. Trans. Am. Soc. Agric. Eng. 50 (3), 875–883.

Logsdon, S.D., James, D., 2014. Closed depression topography Harps soil, revisited. Soil Horiz. 55 (2). https://doi.org/10.2136/sh13-11-0025.

Mallarino, A., Sawyer, J.E., Creswell, J., Tidman, M., 2000. Phosphorus basics. www.ipm.iastate.edu/ipm/icm/2008/8-7-2000/pbasics.html, Accessed date: 20 March 2017.

Miller, J.J., Acton, D.F., St. Arnaud, R.J., 1985. The effect of groundwater on soil formation in a morainal landscape in Saskatchewan. Can. J. Soil Sci. 65, 293–307.

Moore, I.D., Burch, G.J., Mackenzie, D.H., 1988. Topographic effects on the distribution of surface soil water and the location of ephemeral gullies. Trans. ASAE 31, 1098–1107.

Nelson, D.W., Sommers, L.E., 1982. Total carbon, organic carbon, and organic matter. Methods of soil analysis. Part 2. Chemical and Microbiological Properties, 2nd ed. A.L. Page (Ed.) ASA, SSSA, Madison, WI, pp. 539–579.

Olsen, S.R., Cole, R.V., Watanabe, F.S., Dean, L.A., 1954. Estimation of available phosphorus in soils by extraction with sodium bicarbonate. p. 1–19. Circular No. 939. U.S. Dep. of Agric, Washington, D.C.

Pärn, J., Pinay, G., Mander, U., 2012. Indicators of nutrient transport from agricultural catchments under temperate climate: a review. Ecol. Indic. 22:4–15. https://doi.org/10.1016/j.ecolind.2011.10.002.

Priyashantha, K.R.S., Maule, C.P., Elliot, J.A., 2007. Influence of slope position and hog manure injection on fall soil P and N distribution in an undulating landscape. Trans. ASABE 50 (1), 45–52.

Ritchie, J.C., McCarty, G.W., Venteris, E.R., Kaspar, T.C., 2007. Soil and soil organic carbon redistribution on the landscape. Geomorphology 89:163–171. https://doi.org/10.1016/j.geomorph.2006.07.021.

Robertson, W.D., Schiff, S.L., 2008. Persistent elevated nitrate in a riparian zone aquifer. J. Environ. Qual. 37:669–679. https://doi.org/10.2134/jeq2007.0102.

Sacchi, E., Acutis, N., Bartoli, M., Brenna, S., Delconte, C.A., Laini, A., Pennisi, M., 2013. Origin and fate of nitrates in groundwater from the central Po plain: insights from isotopic investigations. Appl. Geochem. 34:164–180. https://doi.org/10.1016/j.apgeochem.2013.03.008.

Sherrod, L.A., Dunn, G., Peterson, G.A., Kolberg, R.I., 2002. Inorganic carbon analysis by modified pressure calcimeter method. Soil Sci. Soc. Am. J. 66:299–305. https://doi.org/10.2136/sssaj2002.2990.

Soil Survey Staff, 2014. Official Soil Series Descriptions. USDA-NRCS. http://soils.usda.gov/technical/classification/osd/index.html, Accessed date: 15 August 2014.

Staricka, J.A., Allmaras, R.R., Nelson, W.W., 1991. Spatial variation of crop residue incorporated by tillage. Soil Sci. Soc. Am. J. 55, 1668–1674.

Thien, S.J., 1979. A flow diagram for teaching soil texture by feel analysis. J. Agron. Educ. 8, 54–55.

Tomer, M.D., Moorman, T.B., Rossi, C.G., 2008. Assessment of the Iowa River's South Fork watershed: part 1. Water quality. J. Soil Water Conserv. 63 (6):360–370. https://doi.org/10.2489/jswc.63.3.360.

Tomer, M.D., Moorman, T.B., Kovar, J.L., Cole, K.J., Nichols, D.J., 2016. Eleven years of runoff and phosphorus losses from two fields with and without manure application, Iowa, USA. Agric. Water Manag. 168:104–111. https://doi.org/10.1016/j.agwat.2016.01.011.

Turkeltaub, T., Kurtzman, D., Dahan, O., 2016. Real-time monitoring of nitrate transport in the deep vadose zone under a crop field — implications for groundwater protection. Hydrol. Earth Syst. Sci. 20 (8):3099–3108. https://doi.org/10.5194/hess-20-3099-2016.

United States Environmental Protection Agency, 1993. Method 365.1, Revision 2.0: Determination of Phosphorus by Semi-automated Colorimetry. US-EPA, Cinncinnati, OH, USA Available online:. https://www.epa.gov/sites/production/files/2015-08/documents/method_365-1_1993.pdf, Accessed date: 3 February 2017.

VandenBygaart, A.J., Broetsch, D., Gregorich, E.G., Lobb, D., 2012. Soil C erosion and burial in cropland. Glob. Chang. Biol. 18:1441–1452. https://doi.org/10.1111/j1365-2486.2011.02604.x.

Wood, E.D., Armstrong, F.A.J., Richards, F.A., 1967. Determination of nitrate in seawater by cadmium-copper reduction to nitrite. J. Mar. Biol. Assoc. UK 47 (1), 23–31.